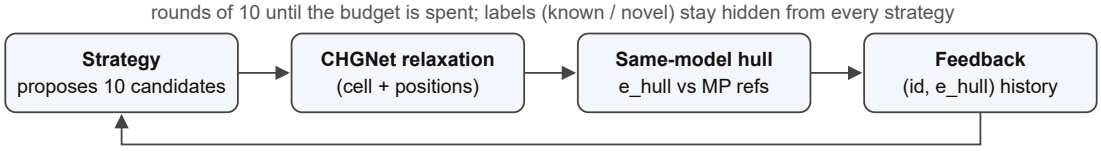
Budgeted Discovery Agent: which Li-halide solid electrolytes are worth simulating?

Team: [add names] · OSI Hackathon 2026 · code: [repo link] · video: [video link]

Problem

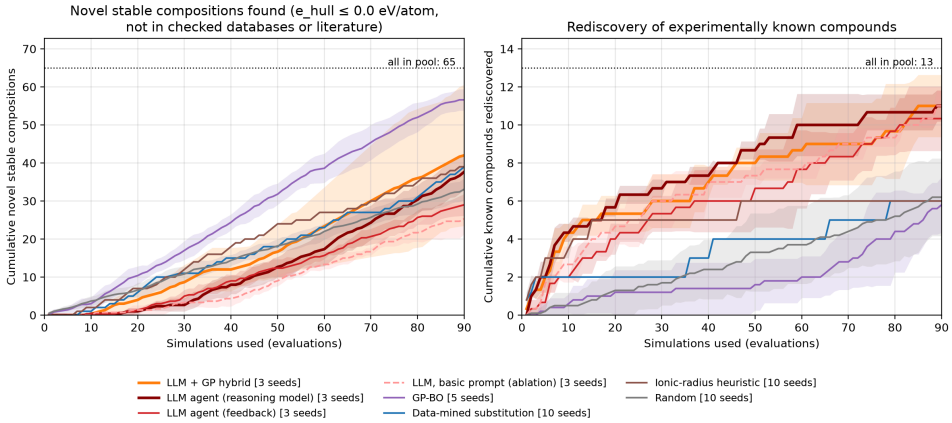
Li3MX6 / Li2MX6 halides (Li3YCl6, Li3InCl6, Li2ZrCl6) are leading solid electrolytes for all-solid-state batteries. Metal, halide and ordering choices make a large space and each candidate costs a simulation, so **deciding what to simulate** is the bottleneck. We built 236 candidates (123 compositions) by substituting Materials Project parents: 22 trivalent and 5 tetravalent metals, Cl/Br, ordered mixed-halide and mixed-metal variants.

Method



Every evaluation is a CHGNet relaxation scored by its energy above a convex hull of 513 MP reference phases relaxed with the same potential (validation: MAE 14 meV/atom vs DFT over 399 compounds; all DFT-stable ternaries within 0.03 eV/atom). Hit = e_hull ≤ 0.0 eV/atom (on or below the hull). The LLM agents (gpt-4.1-mini; reasoning arm gpt-5-mini) see all unevaluated candidates and the full (id, description, e_hull) history and return 10 picks with a chemical reason each. Baselines: random, ICSD substitution probabilities (Hautier et al.), ionic-radius similarity, GP-BO on Magpie features; all with the same budget (90 of 236).

Result



With 90 simulations from a pool of 236 candidates (65 novel stable compositions hidden in it), GP-BO found the most novel stable compositions, 56.6 ± 2.8 versus 33.1 ± 2.8 for random: random search needs 1.7x as many simulations to match it (1.5x on all stable compositions), and the data-mined substitution baseline reaches 1.2x. The best LLM arm (LLM + GP hybrid) reached 1.3x on novel and 1.3x on all stable compositions. LLM agents were the fastest at rediscovering experimentally known compounds (LLM, basic prompt: 11.0 ± 0.8 of 13 vs 6.2 ± 2.0 for random), consistent with literature memory; the same prior pulls them toward known chemistry and away from novel compositions.

strategy (@90 sims)	seeds	stable found	novel found	known rediscovered	speed-up vs random (stable / novel)
LLM + GP hybrid	3	58.0 ± 16.5	42.0 ± 18.5	11.0 ± 1.6	1.3x / 1.3x
LLM agent (reasoning model)	3	54.3 ± 3.9	37.7 ± 4.7	11.0 ± 0.8	1.3x / 1.1x
LLM agent	3	43.7 ± 1.7	29.0 ± 2.8	10.3 ± 1.7	1.0x / 0.9x
LLM, basic prompt	3	41.0 ± 2.9	25.3 ± 4.0	11.0 ± 0.8	0.9x / 0.8x
GP-BO	5	65.8 ± 4.6	56.6 ± 2.8	5.8 ± 1.5	1.5x / 1.7x
Data-mined substitution	10	48.0 ± 0.0	39.0 ± 0.0	6.0 ± 0.0	1.1x / 1.2x
Ionic-radius heuristic	10	49.0 ± 0.0	39.0 ± 0.0	6.0 ± 0.0	1.1x / 1.2x
Random	10	43.2 ± 2.9	33.1 ± 2.8	6.2 ± 2.0	1.0x / 1.0x

New candidates (CHGNet + MACE-MP-0 cross-check; 15/15 agreed within 0.05 eV/atom)

formula	prototype	CHGNet e_hull	MACE e_hull	agreed	databases	literature
Li2CeBr6	Li2ZrCl6-type	-0.133	-0.032	yes	absent*	none found
Li3YbBr6	Li3InCl6-type	-0.070	-0.003	yes	absent*	none found
Li3Ho0.5Yb0.5Cl6	Li3InCl6-type	-0.060	-0.037	yes	absent*	none found
Li3Yb0.5Dy0.5Cl6	Li3YCl6-type	-0.054	-0.017	yes	absent*	none found
Li2ZrBr6	Li2ZrCl6-type	-0.032	+0.015	yes	absent*	none found

Validation and limitations

ML potentials approximate DFT (CHGNet reference MAE 14 meV/atom; MACE 21 meV/atom); stability margins here are a few meV/atom, comparable to that error, so hits are predictions, not proof. Yb compounds come out most stable with CHGNet but much less so with MACE (Yb reference energies are known to be delicate in MP-trained models); Li2CeX6 needs Ce4+, which is not guaranteed to survive. Ordered approximants of partly disordered phases; 0 K thermodynamics only (no ionic conductivity or electrochemical window); campaigns replay precomputed evaluations. The brief's 0.03 eV/atom hit threshold makes 92% of the pool 'hits', so we used e_hull <= 0 (0.03/0.05 in summary.json). *'Novel' = not found in MP, GNoME, JARVIS, OQMD (partly reachable) and literature searches; the Alexandria server was unreachable during the run and is reported as unchecked.